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Light-touch forecasting: A novel method to combine human judgment with statistical algorithms

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ABSTRACT

Forecast adjustments are an indispensable component of the forecasting process, but what is the most effective and efficient method to make these adjustments? Understanding how to effectively blend human forecast adjustments with statistical methods is of great importance as, even with the increased possibilities of AI, we can still not include all information available to a human planner. We address this issue by developing a method that builds on literature showing that some adjustments are consistently (in)accurate. More specifically, in two extensive case studies containing more than 3.5 million forecasting decisions, we confirm that planners are accurate in adjusting a statistical forecast in the right direction and determining the magnitude of downward adjustments, and they are inaccurate in determining the magnitude of upward adjustments. Leveraging these results, we introduce a novel method called *light-touch forecasting,* which attains performance levels similar to those of more traditional forecasting methods while minimizing the involvement of human planners. Furthermore, an online experiment shows an efficiency gain of 38% in terms of time spent on planning compared to traditional judgmental forecasting. We thus optimize the forecasting process by using the strengths of planners while avoiding their weaknesses.

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1. Introduction

Processes in businesses and everyday life are digitizing rapidly. Technological advancements, including sensor technology and artificial intelligence (AI), enable the Industry 4.0 revolution (Makridakis et al., 2022; Olsen & Tomlin, 2020). Organizations need to transform their current working methods to accommodate these technological advancements and remain competitive. More specifically, a report by McKinsey (2018) on supply chain planning argues that for companies to stay ahead of the competition, it is crucial to differentiate between processes that can be fully automated and processes that still need human attention. Even when AI or other automated systems outperform humans, performance may improve considerably when humans and systems collaborate properly. For example, instead of automating complete tasks, it may be better to automate only some task instances (Autor et al., 2003; Fügener et al., 2020). The question is for which task does this hold, and how should the collaboration be designed? For most tasks in operations management, this question is still open. According to McKinsey (2018), short-term demand forecasting for stock-keeping units is a task that can already be fully automated (i.e., without the need for human attention or intervention). We challenge this notion by proposing a method combining statistical forecasts and human involvement.

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In the current way of forecasting the demand for products, the large amount of (historical) data often necessitates companies to rely on statistical algorithms to generate forecasts, e.g., time-series methods or black-box machine learning methods (Goodwin & Fildes, 1999; Harvey, 1995; Schaer et al., 2022). Demand planners typically review and adjust system-generated forecasts to include information the system lacked when generating its demand forecast (Fildes et al., 2023). This non-historical data can be anything perceived to be relevant in influencing future demand levels, such as upcoming promotional campaigns, uncertainties in supply chains, geopolitical events, or pandemics. When taking into account this non-historical information, demand planners often propose adjustments that are too optimistic (Das et al., 1998; Fildes et al., 2009; McNees, 1990), see patterns in random fluctuations (Fildes et al., 2009; Harvey, 1995; O'Connor et al., 1993), and excessively anchor on the value of the last demand period in making new demand projections (Harvey, 2007; Lawrence & O'Connor, 1993; Lawrence et al., 2000). On the other hand, research also indicates that planners can improve forecast accuracy. That is, planners can select and adjust the most inadequate system-generated forecasts (Mathews & Diamantopoulos, 1990), and they are relatively accurate in correcting the statistical forecast in the right direction, specifically in determining that the statistical forecast should be lower (Fildes et al., 2009; Petropoulos et al., 2016). Given that (1) judgmental adjustments can increase or decrease forecast accuracy and (2) these adjustments can be costly in terms of time and effort, finding better ways to combine statistical forecasts with human judgment is of great interest to academics and forecasters (Fildes et al., 2009; Khosrowabadi et al., 2022; Lawrence et al., 2006; Perera et al., 2019).

Research on the effective and efficient integration of human judgments with statistical algorithms is somewhat limited, as most research focuses on identifying those situations in which planners can add value to the forecasting process. One example where human judgment is integrated with statistical algorithms is the work of Franses and Legerstee (2013). They use the decisions the human planner made in the past as an additional variable in the forecasting equation, which is useful when the statistical model does not perform well. Another example is the work of Blattberg and Hoch (1990), in which taking the average of a planner's forecast and a statistical forecast is proposed to generate one's final forecast. Yet, it is uncommon for companies to use this method in this exact format, as planners are typically aware of the statistical forecast and, as a result, tend to anchor on this value when proposing a new forecast (Fildes et al., 2009). Likewise, a planner may be tempted to adapt her forecast, knowing that the final forecast will be an average of that forecast and the statistical forecast. Franses and Legerstee (2013) and Blattberg and Hoch (1990) propose a new forecasting method combining human judgment and statistical algorithms. Still, neither of them capitalizes on the strengths of human planners while avoiding their mistakes.

Building on earlier findings in the literature on judgmental forecasting, we develop a method in which we focus on the strengths of human planners, i.e., we let them decide on the *direction* of the forecast adjustment and the *magnitude* of downward adjustments. At the same time, we also consider their predictable mistakes, i.e., the statistical algorithm decides on the *magnitude* of the upward adjustment. We name this method *light-touch forecasting,* and it has three objectives: First, to improve the forecasting accuracy, i.e., reduce the forecasting error. Second, to improve the efficiency of the forecasting task, i.e., reduce the time needed by the human planner. Third, to ensure that the method cannot be easily gamed, according to the method of Blattberg and Hoch (1990), planners can adapt their adjustments, participating in what the method will do to their adjustments in the next step. This is not possible in our method.

To accomplish these objectives, we use data from two large companies, referred to as case studies, to analyze planners' adjustments to statistical forecasts. Each forecast adjustment is compared to the accuracy of the initial statistical forecast and the actual sales of a product. Our results confirm earlier findings from the literature. That is, planners are reasonably good at determining whether to adjust a statistical forecast upward or downward. Planners are consistently outperformed by the forecasting algorithm when making upward adjustments, while planners improve the forecasting accuracy when making downward adjustments. By testing the light-touch forecasting method in two different industries and over various time lags, i.e., six, four, two, and zero months before the actual demand, we consistently show that using the strengths of planners (while avoiding their weaknesses) can achieve similar performance levels compared to more traditional forecasting methods, while decreasing the time needed to make adjustments. This finding aligns with earlier research by Goodwin et al. (2011).

We next performed an online experiment with 484 planning professionals and participants from other fields to test whether planners behave differently when confronted with light-touch forecasting and to measure the time needed to perform the forecasting task. We observe comparable usage behaviors in both the traditional and light-touch forecasting methods, suggesting that the adoption of our method may indeed yield substantial performance enhancements, as exemplified in the case studies. Furthermore, our method shows large efficiency gains as light-touch forecasting is less time-consuming than current forecasting methods in which planners are asked to overwrite all statistical forecasts manually; i.e., light-touch forecasting is 38% faster than traditional judgmental forecasting. This confirms the second objective of light-touch forecasting. It also shows that forecasting performance remains the same if a planner is restricted in their judgment (objective 3).

To summarize, our contribution is as follows. We aim to enhance forecasting performance and efficiency by capitalizing on the strengths of human planners while mitigating their weaknesses by developing a novel approach called 'light-touch forecasting.' This method ensures the inclusion of human expertise on current or future market trends into a forecast, which may be overlooked by the system's analysis of historical sales patterns.

Simultaneously, it works to prevent or mitigate any errors made by human input. To showcase its effectiveness, we analyze two separate case studies from distinct industries to illustrate that our forecasting method attains performance levels similar to those of current, more traditional forecasting methods. Finally, we conducted an online experiment to demonstrate that individuals show similar usage behaviors when using the light-touch forecasting method, completing the task significantly faster. This multi-method approach is not commonly used in the judgmental forecasting literature, and the results have practical implications as implementing the light-touch forecasting method in companies can lead to a similar forecasting accuracy while decreasing the time needed to perform the actual forecasting task.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature on judgmental forecasting and presents its findings. Section 3 describes our light-touch forecasting method, building on those findings. Section 4 describes the setting of our empirical case studies, while the results are given in Section 5. We present the setup and results of our online experiment in Section 6. Section 7 concludes the paper and offers directions for future research.

2. Literature review

In the domain of forecasting, a considerable amount of research has focused on understanding how judgmental adjustments impact the accuracy of a forecast. In what follows, we first describe companies' typical (judgmental) forecasting procedure when estimating future sales. Next, we give a detailed description of findings from the literature.

2.1. Planners make many adjustments and do this more accurately when adjusting downwards

Statistical forecasting analyzes historical demand data to predict future product demand (Hyndman & Athanasopoulos, 2018). The most common method of forecasting demand starts with a statistical forecast based on historical demands, which is accepted or adjusted by demand planners to take into account events that are unknown to the statistical model (e.g., Davydenko & Fildes, 2013; Fildes et al., 2009). In the context of demand forecasting, judgmental adjustments refer to the process in which demand planners revise and overwrite a system-generated forecast by manually entering a new expected demand value into the system (Davydenko & Fildes, 2013; Fildes et al., 2009; Lawrence et al., 2006). Describing the impact of these types of adjustments in predicting future demand is among the most active research areas in the field of judgmental forecasting (e.g., Blattberg & Hoch, 1990; Fildes & Goodwin, 2007; Fildes et al., 2023, 2009; Franses & Legerstee, 2009; Goodwin & Fildes, 1999; Hogarth & Makridakis, 1981). These adjustments have generated much interest because of the consistent finding that judgmental adjustments are very common. For example, by analyzing more than 60,000 forecasts in various companies, Fildes et al. (2009) found that planners revise between 60 and 95 percent of the statistical forecasts (see Franses and Legerstee (2009) for similar findings).

Although the judgmental forecasting procedure seems rational, the evidence of whether these adjustments improve the forecast accuracy is mixed. That is, while some studies suggest that these judgmental interventions improve accuracy (e.g., Fildes et al., 2009; Lawrence et al., 1985, 1986, 2000; Petropoulos et al., 2016), other studies indicate that these adjustments can seriously harm the accuracy of a forecast (e.g., Carbone & Gorr, 1985; Fildes et al., 2019; Sanders, 1992, as planners are subject to various biases (Eroglu & Sanders, 2022; Karelse, 2021)). For example, planners are consistently too optimistic about the future (Das et al., 1998; Eroglu & Croxton, 2010; Fildes et al., 2009; Khosrowabadi et al., 2022). That is, due to optimism, planners are much more likely to adjust a forecast in an upward direction, although this optimism often proves to be unwarranted. A systematic review of the human factor in forecasting can be found in the article by Perera et al. (2019).

Given that human planners systematically adjust statistical forecasts (in the upward direction), we discern the first finding from the literature:

FINDING 1. *Human planners adjust statistical forecasts regularly. Due to optimism bias, these adjustments are mostly in an upward direction.*

Similarly, these upward adjustments tend to be excessively large, leading to a systematic increase in forecast error. However, although upward adjustments typically decrease the accuracy of a demand forecast, research also demonstrates that downward adjustments usually improve the quality of a forecast (Fildes et al., 2009), an observation that prompted Alvarado-Valencia et al. (2017) to suggest that only upward adjustments should be evaluated. The finding that downward adjustments are more accurate is confirmed in the research of Khosrowabadi et al. (2022). Given that human planners increase the forecast error when making upward adjustments while decreasing it when making downward adjustments, we discern the second finding from the literature:

FINDING 2. *In general, upward adjustments by planners increase the forecast error, while downward adjustments decrease the forecast error.*

2.2. Planners are accurate in determining the adjustment direction

Research repeatedly shows that humans are predictably inaccurate when executing some types of tasks while they are highly accurate in executing others (e.g., Campbell, 1988; Dawes, 1979; Meehl, 1954). The key feature in determining whether people perform predictably (un)satisfactorily seems to center on the task's inherent complexity (Campbell, 1988). While people are relatively accurate in executing simple estimation tasks (e.g., determining which of two buildings is tallest), they have difficulties in more complex estimation tasks (e.g., determining how much taller the tallest building is). To translate this to demand forecasting is difficult, as the components of the forecasting task (i.e., deciding on the adjustment direction and magnitude) involve large

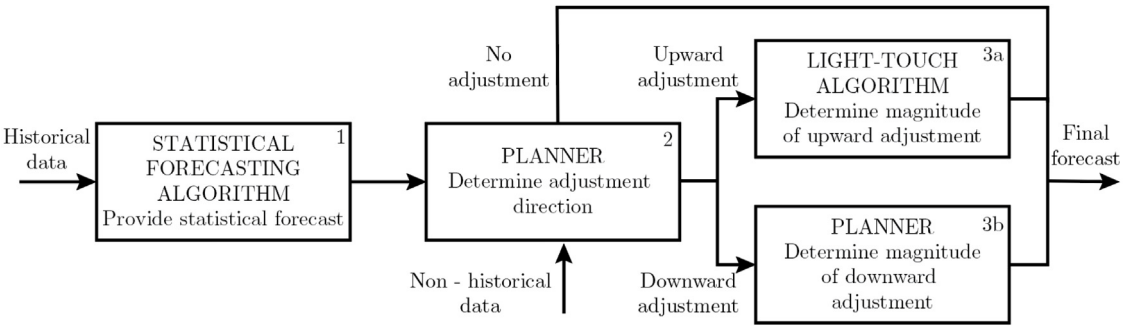


Fig. 1. Light-touch forecasting method.

amounts of data and multiple stakeholders. Judgmentally adjusting a system's generated forecast requires a decision-maker to evaluate and integrate multiple data sources and decide on the likelihood of an event occurring. The number of possible outcomes can lead to an information overload for the decision-maker, which increases when the complexity of the task increases (Campbell, 1988). The way humans respond to increased task complexity also has consequences for forecasting. In demand forecasting, deciding whether the statistical forecast should go up or down is a relatively simple task, as it involves only two options to evaluate. When deciding to adjust a statistical forecast, planners often respond to available market information, impacting the demand for a particular product in a specific direction (e.g., upcoming promotional campaigns are likely to positively influence demand while new competitors entering the market harm demand levels). Consequently, planners are likely to adjust in the right direction. As shown by Fildes et al. (2009), for both companies in the authors' research, between 68% and 63% of the forecasts were adjusted in the correct direction (see Petropoulos et al. (2016) for similar results). Fildes et al. (2009) furthermore split the correctly directed adjustments into upward and downward adjustments, and they conclude that for upward adjustments, between 65.6% and 49.4% is in the correct direction, and for downward adjustments, this percentage is much higher, i.e., between 79.5% and 71.7%. Therefore, the third finding from the literature is:

FINDING 3. *Overall, planners are accurate in adjusting statistical forecasts in the right direction. When adjusting a forecast, planners accurately decide whether the statistical forecast should go up or down. This effect is most pronounced when adjusting a statistical forecast downwards.*

3. Light-touch forecasting

To combine the strengths of planners and statistical forecasting algorithms, we propose a 'light-touch forecasting' method that builds on the findings from the judgmental forecasting literature. Fig. 1 shows the three steps of the method and the agent performing each step: a statistical forecasting algorithm, the light-touch algorithm, or a human planner.

First, a statistical forecast is generated by a forecasting algorithm using historical data. Depending on the company, different forecasting algorithms can be used (e.g., moving average, Croston's method, seasonal naïve). Second, the planner decides whether to accept or adjust the statistical forecast using non-historical data that the planner has commonly referred to as market intelligence or domain knowledge (Lawrence et al., 2006). In the former case, we have the final forecast. In the latter case, the planner also decides on the adjustment direction (i.e., whether the statistical forecast should be adjusted upwards or downwards). Third, when a planner indicates that the statistical forecast should be adjusted upwards, the algorithm adds x% to the statistical forecast to determine the final forecast (see below on how x is set). This new forecast is rounded to the nearest integer. When a planner indicates that the statistical forecast should be adjusted downwards, (s)he determines the magnitude of the adjustment (i.e., how many units the statistical forecast should be lowered).

To be more precise on the third step and the final forecast, we define them in mathematical terms. Let p and t denote the product and the time period for which we want to forecast demand (for this explanation, it is not relevant to know at which moment in time the forecast is performed). Further, let $F^{S}_{t,p}$, $F^{H}_{t,p}$, and $F^{F}_{t,p}$ denote the statistical forecast, the forecast of the planner, and the final forecast for period t and product p, respectively. The standard deviation of these is denoted with d. The final forecast is now defined as

$$F^{F}_{t,p} = \begin{cases} F^{H}_{t,p} & \text{if downward adjustment;} \\ F^{S}_{t,p} & \text{if no adjustment;} \\ F^{S}_{t,p}(1 + \frac{x}{100}) & \text{if upward adjustment.} \end{cases} \quad (1)$$

To explain how to determine x for upward adjustments, we first define a training set of periods T and products P (we give more details on these sets in Section 4.3). We consider each integer value $x' \in -100, -99, \ldots, 99, 100$; we allow x' to be negative to take into account that planners can wrongly decide on the adjustment direction. For each period $t \in T$ and product $p \in P$ for which a planner proposed to adjust upwards, i.e., $F^{H}_{t,p} > F^{S}_{t,p}$, we calculate $F^{S}_{t,p}(1 + \frac{x'}{100})$. We then calculate the mean error that results over all periods and products. We set x to that value x' that gives the lowest mean error (we provide more details on calculating the error term in Section 4.2).

In conclusion, we propose a method, light-touch forecasting, for making forecast adjustments that capitalizes on the strengths of planners while avoiding their weaknesses. As such, we formulate a novel adjustment method that effectively blends human judgment with system-generated forecasts to increase the overall forecast accuracy. Building on earlier findings, as described in Section 2, we argue that the forecasting error can be significantly lowered when planners decide on the direction of an adjustment and the magnitude of downward adjustments. In contrast, the forecasting algorithm decides on the magnitude of upward adjustments. More specifically, we predict:

Hypothesis 1. The forecast error of the light-touch forecasting method is lower than the error of both the statistical forecasting algorithm and the planner.

We furthermore predict that the time it takes to execute the forecasting task is reduced significantly, i.e., the efficiency is improved significantly, as a planner has to make fewer decisions. We hypothesize:

Hypothesis 2. The efficiency of executing the forecasting task is higher using the light-touch forecasting method than the current (all-touch) forecasting method.

Next to performance and efficiency, we evaluate the possibility of using the gaming method, meaning that a planner enters a higher or lower forecast to still get their desired result after applying the light-touch or Blattberg and Hoch method. The light-touch forecasting method does not allow the planner to circumvent the proposed method. This is a potential issue for the method of Blattberg and Hoch as it is very intuitive, and a planner can easily circumvent the method by, for example, doubling its adjustment.

4. Research setting

4.1. Case descriptions

We have collected data from two large, multinational companies for which Company A is active in the consumer goods industry. At the same time, Company B is active in the construction and building maintenance industry. Both companies met with several planners and managers to better understand the forecasting process and planners' incentives. Planners in Company A forecast (retail) demand three months before the actual demand period. In contrast, planners in Company B forecast (end-user) demand six, four, two, and zero months before the actual demand period (i.e., a rolling forecast). Planners gather additional non-historical information to include relevant market information when making adjustments. Planners of Company A discuss internally with their sales managers, portfolio managers, and marketing managers what further information is available for a specific product. They then integrate this market knowledge into the statistical forecast. This makes the statistical forecast a group effort instead of an individual effort by the planner. Planners of Company B communicate regularly with country managers to obtain country-specific non-historical information that they use when making forecast adjustments. In both companies, each data point contains the original statistical forecast, the adjustment proposed by the planner, and the actual demand for a stock-keeping unit (SKU) in a particular country. Planners receive an initial statistical forecast that can either be accepted or adjusted. When a planner chooses to adjust the statistical forecast, a new value is entered into the system (Company A), or the amount that should be added or subtracted from the statistical forecast is entered (Company B). In Company A, planners are not incentivized based on the accuracy of forecasts, while in Company B, planners obtain a financial reward when the overall group obtains excellent results. In both companies, the (final) forecasts are subsequently used by another department to control inventories.

4.2. Error term

Various measures can be used to examine the accuracy of the forecasts obtained. There is an ongoing debate on which error measure to use in the literature, as no method is perfect and applicable to all contexts (Armstrong & Fildes, 1995). We use the symmetric mean absolute percentage error (sMAPE), the median absolute percentage error (MdAPE), and the mean absolute error (MAE). The sMAPE and MdAPE scale the difference between the actual demand and the forecast relative to the actual demand. Moreover, MdAPE and sMAPE can handle outlier values well. The mean absolute error is easily interpretable, making it easy to compare the performance of the different forecasting methods. In the remainder of the paper, we will present the sMAPE, MdAPE, and MAE results.

To formally define the sMAPE, we first introduce $y \in \{S, H, F, B\}$ to be the forecast entity, with S denoting the statistical forecast algorithm, H the human planner, F the light-touch forecasting method, and B the method of Blattberg and Hoch (1990). Using the notation introduced in Section 3, we further introduce $F^{B}_{t,p}$ as the forecast of the latter method for period $t \in T$ of product $p \in P$, with P being the set of products:

$$F^{B}_{t,p} = \frac{F^{S}_{t,p} + F^{H}_{t,p}}{2}.$$

Let $A_{t,p}$ denote the actual demand in period $t \in T$ of product $p \in P$. The error made by forecast entity y over the complete set of products P and periods T can now be defined as

$$E^{y} = \frac{1}{\overline{\overline{P}}} \frac{1}{\overline{\overline{T}}} \sum_{p \in P} \sum_{t \in T} \frac{|A_{t,p} - F^{y}_{t,p}|}{((A_{t,p} + F^{y}_{t,p})/2)}.$$

When discussing the results, we are sometimes only interested in the error for those forecasts that have been adjusted in the same direction (or not at all). To that end, we introduce $w \in$ U, N, D to denote the adjustment direction of the human planner, with U representing an upward adjustment, N representing no adjustment, and D representing a downward adjustment. We next define for a product $p \in P$ the set T^{w}_{p} to be the set of periods in

Table 1
Number of data points left after each data cleaning step.

Company			Training set				Validation set		
			Initial	Missing data	Null forecast	Outliers	Initial	Missing data	Null forecast
sMAPE	A		221,010	46,446	41,693	41,693	41,126	30,753	27,760
	B	Lag 6	1,131,871	580,775	486,688	486,688	435,335	305,228	289,732
		Lag 4	1,131,871	686,095	592,517	592,517	435,335	300,676	284,781
		Lag 2	1,131,871	738,859	700,708	700,708	435,335	296,162	281,615
		Lag 0	113,1,871	728,546	699,989	699,689	435,335	292,175	231,857
MdAPE	A		221,010	46,446	41,665	41,563	41,126	30,753	27,760
	B	Lag 6	1,131,871	580,775	392,211	391,772	435,335	305,228	234,555
		Lag 4	1,131,871	686,095	485,374	484,752	435,335	300,676	233,366
		Lag 2	1,131,871	738,859	577,898	577,036	435,335	296,162	251,685
		Lag 0	113,1,871	728,546	580,238	578,524	435,335	292,175	231,857
MAE	A		221,010	46,352	41,612	41,387	41,126	30,847	27,882
	B	Lag 6	11,331,871	527,527	486,688	485,640	435,335	304,764	289,732
		Lag 4	11,331,871	633,702	592,517	591,213	435,335	300,096	284,781
		Lag 2	11,331,871	736,580	700,708	699,199	435,335	295,443	281,615
		Lag 0	11,331,871	726,307	699,689	698,257	435,335	291,389	279,070

which the human planner adjusted the statistical forecast in the direction w

$$T_p^w = \begin{cases} \{t \in T | F_{t,p}^{\mathrm{H}} < F_{t,p}^{\mathrm{S}}\} & \text{if } w = \mathrm{D}; \\ \{t \in T | F_{t,p}^{\mathrm{H}} = F_{t,p}^{\mathrm{S}}\} & \text{if } w = \mathrm{N}; \\ \{t \in T | F_{t,p}^{\mathrm{H}} > F_{t,p}^{\mathrm{S}}\} & \text{if } w = \mathrm{U}. \end{cases}$$

The definitions of the MdAPE and MAE are given in Appendix A.

4.3. Analysis method

The data set was split into two sets: a training set and a validation set. This is to test the light-touch forecasting method. The training set was used to parameterize the forecasting method, and the validation set was used to determine its predictive value (Tashman, 2000). We used 80% of our data set as a training set and the remainder for validation (Hyndman, 2014). The training set contained June 2015 to March 2017 for Company A and July 2017 to July 2018 for Company B, while the validation data set contained the months April 2017 to November 2017 for Company A and August 2018 to December 2018 for Company B.

Following Fildes et al. (2009), we excluded incomplete data points, i.e., data points with missing information on either the actual demand, statistical forecast, or an adjusted forecast of a planner, and data points with zero as forecast by a planner, i.e., $F_{t,p}^{H} = 0$. Including those forecasts in the data where no sales are expected would favor the accuracy of human planners and the light-touch forecasting method compared to the statistical algorithm; they would have an error of zero. To calculate the median absolute percentage error (as depicted in Appendix A), we deleted actual demands of zero, as dividing by zero is undefined. Finally, outliers were removed from the training set based on z-scores (Field, 2009, p. 102). When the absolute z-scores of both the statistical forecast error and the human forecast error were larger than three, we classified the data point as an outlier, as this indicated a special case in which the available data and information were not sufficient to be able to predict what happened in that specific month and SKU-location combination. Outliers were not deleted in the validation set, as that was used to simulate what would have happened in practice when outliers were unclear beforehand. An overview of the number of data points remaining after the various data-cleaning steps is given in Table 1.

5. Results

In this section, we first test whether the findings from Section 2 are also observed in the two companies. After that, we test the performance of the light-touch forecasting method and compare it to the performance of the statistical forecasting algorithm, the human planner, and the method of Blattberg and Hoch (1990).

5.1. Evaluation of findings from the literature

Table 2 shows some descriptive measures of the planners' adjustments, including their standard deviation, denoted with d. In both companies, numerous statistical forecasts were adjusted, and most adjustments were in the upward direction, confirming Finding 1. When a planner adjusted the statistical forecast in a downward direction, the symmetric mean absolute percentage error of the planner ($E^{\mathrm{H,D}}$) was lower compared to the error of the initial statistical forecast ($E^{\mathrm{S,D}}$) for Company A. The errors are very similar to each other for Company B. However, for upward adjustments, this effect reversed for Company A: the error of the planner ($E^{\mathrm{H,U}}$) was higher (or very much similar to) than the error of the statistical forecast ($E^{\mathrm{S,U}}$). For Company B, the errors remain very similar to each other. The results of Company A confirm Finding 2, whereas the results of Company B do not confirm this finding. Similarly, adjustments in both companies were often in the right direction. This effect was most pronounced for downward adjustments, thus confirming Finding 3. Similar results are found for the two other error terms.

Table 2
Descriptive measures for adjustments at Companies A and B.

	Company	A	B			
			Lag 6	Lag 4	Lag 2	Lag 0
Finding 1	Sample size (N)	41,693	486,688	592,517	700,708	699,689
	Forecasts adjusted [%]	98.8	21.0	32.5	40.6	44.5
	- Downward [%]	39.9	4.2	6.7	8.1	9.1
	- Upward [%]	59.0	16.8	25.8	32.4	35.4
Finding 2	$E^{S,D}$	0.73	0.73	0.71	0.70	0.68
	($d^{S,D}$)	0.53	0.62	0.61	0.62	0.61
	$E^{H,D}$	0.64	0.74	0.72	0.72	0.70
	($d^{H,D}$)	0.50	0.63	0.62	0.62	0.62
	$E^{S,U}$	0.65	0.78	0.74	0.72	0.71
	($d^{S,U}$)	0.49	0.63	0.63	0.63	0.62
	$E^{H,U}$	0.68	0.76	0.73	0.73	0.71
	($d^{H,U}$)	0.52	0.63	0.63	0.62	0.62
Finding 3	Correct direction [%]	63.3	55.6	53.4	52.2	52.0
	- Correct out of downward [%]	77.7	61.5	65.5	63.4	62.3
	- Correct out of upward [%]	53.6	51.3	50.2	49.4	49.4

5.2. Performance of the light-touch forecasting method

As discussed in Section 3, the light-touch forecasting method adds x% to the statistical forecast when a planner indicates that a statistical forecast should be higher. Table 3 shows that x is relatively low and constant, varying between 10 and 13. Importantly, Table 3 depicts the forecasting performance of the statistical algorithm, the human planner, the light-touch forecasting method, and the method of Blattberg and Hoch (1990). The light-touch forecasting method performs similarly to the statistical forecasting algorithm and the human planner (i.e., $p > .05$). At the same time, it is more time efficient, as a human planner must perform a smaller part of the adjustments. The method of Blattberg and Hoch (1990) is similar to the human planner and statistical forecast (i.e., $p > .05$). The performance of the Blattberg-Hoch method is very similar to the light-touch method. We observe that the MdAPE and sMAPE results are very similar. For the MAE, the light-touch forecasting method performs better than the planner and statistical models. At the same time, its performance is still comparable to the performance of the method of Blattberg and Hoch. We extend the light-touch forecasting method by differentiating x based on volume, i.e., average sales, and the coefficient of variation of the historical demand. Differentiating based on product volume and coefficient of variation does not lead to a significantly lower forecasting error. This contradicts the results of Khosrowabadi et al. (2022); they find that product-related variables are important to consider when evaluating the performance of adjustments.

Next to the comparison in forecast error, we also evaluate the forecast added value, which is the percentage of forecasts improving in terms of error when using the light-touch forecasting method. We only consider adjusted forecasts by comparing the light-touch forecasting method with the statistical method, planner, and method of Blattberg and Hoch (1990). For all comparisons at both companies and all lags, Table 4 shows that the percentage of forecasts improving in the light-touch forecasting method is higher than that of forecasts deteriorating in terms of error for the statistical forecast and the planner. Comparing the light-touch method with the method of Blattberg and Hoch shows that adjustments positively affect performance for Company A as more forecasts are improved than deteriorated. This is reversed for Company B. In terms of MAE, the light-touch forecasting method improves a higher number of forecasts than it deteriorates for all companies and lags. For the MdAPE, the light-touch forecasting method improves a significant number of SKUs for Company A, whereas this is not the case for Company B.

6. Experiment on light-touch forecasting

Restricting someone's decision freedom, as proposed by the light-touch forecasting method, can lead to reactance, i.e., a person adopting behaviors or views different from what is intended (i.e., Steindl et al., 2015). We have already explained in Section 3 that planners cannot game the light-touch forecasting method. Our experiment aims to determine whether people would still behave differently, e.g., deciding to adjust more often in the downward direction, when using the (restrictive) light-touch forecasting method. By doing so, we aim to determine the applicability of our method in practice. In addition, the light-touch forecasting method should be less time-consuming, given that planners do not need to manually insert a forecast when they adapt the statistical forecast upwards. We, therefore, expect that participants will perform the forecasting task significantly faster when using the light-touch forecasting method, providing us with an indication of the potential time savings of this method in practice.

In what follows, we describe the task participants performed in the experiment, then present and discuss its results. Note that when presenting the results, we do not discuss the error of participants' decisions, as too little information was provided to them to make adequate adjustments in terms of magnitude. Instead, we only focus on the number and type of adjustments participants made across conditions.

Table 3
Error statistics of Company A and B.

Company			sMAPE			MdAPE			MAE		
			Mean	d	x	Mean	d	x	Mean	d	x
A		E^S	0.68	0.51		0.54	33.10		616.6	2,574.1	
		E^H	0.68	0.52		0.58	33.10		659.4	2,692.2	
		E^F	0.65	0.51	10	0.51	24.74	−19	551.1	2,343.7	5
		E^B	0.65	0.51		0.53	45.36		578.5	2,318.2	
B	Lag 6	E^S	0.87	0.69		0.48	9.94		4,284.9	78,839.0	
		E^H	0.86	0.69		0.48	11.80		4,522.8	84,755.1	
		E^F	0.87	0.69	13	0.48	9.53	−7	4,262.5	78,201.8	3
		E^B	0.86	0.69		0.47	10.58		4,240.6	76,446.3	
	Lag 4	E^S	0.84	0.69		0.47	6.08		4,142.0	74,620.3	
		E^H	0.84	0.69		0.48	7.04		4,301.8	76,614.4	
		E^F	0.82	0.67	13	0.46	5.97	−2	4,086.1	73,296.2	3
		E^B	0.83	0.69		0.46	6.43		4,093.8	72,825.1	
	Lag 2	E^S	0.82	0.68		0.45	6.83		3,937.5	66,327.7	
		E^H	0.82	0.68		0.46	7.54		4,071.4	67,855.3	
		E^F	0.82	0.68	10	0.44	6.71	−4	3,876.0	65,692.4	1
		E^B	0.81	0.68		0.45	7.09		3,876.1	63,505.8	
	Lag 0	E^S	0.55	0.45		0.43	8.92		3,801.4	60,446.5	
		E^H	0.55	0.45		0.44	9.25		3,887.5	65,158.9	
		E^F	0.55	0.45	10	0.43	8.59	−12	3,704.6	59,857.2	2
		E^B	0.55	0.44		0.43	8.98		3,701.1	57,603.7	

Table 4
Forecast added value Company A and B.

	Company		Light-touch forecasting		Human planner		Blattberg and Hoch (1990)	
			Pos.	Neg.	Pos.	Neg.	Pos.	Neg.
sMAPE	A		54.9	43.7	36.7	25.5	53.2	46.0
	B	Lag 6	45.5	35.8	36.0	36.1	39.5	44.3
		Lag 4	45.7	36.9	34.4	33.7	40.6	43.9
		Lag 2	44.3	38.6	34.1	31.0	41.6	43.3
		Lag 0	49.1	46.0	37.6	32.9	47.4	49.4
MdAPE	A		50.0	49.1	37.3	25.4	54.4	45.5
	B	Lag 6	36.1	51.6	42.1	44.7	44.6	55.1
		Lag 4	35.0	42.7	41.0	40.8	46.6	53.1
		Lag 2	39.8	46.6	40.2	38.0	48.0	51.8
		Lag 0	42.7	51.9	37.7	37.0	47.4	52.3
MAE	A		55.6	41.6	39.1	23.2	55.8	43.7
	B	Lag 6	42.6	36.1	48.4	36.8	52.6	46.0
		Lag 4	43.5	36.4	45.9	34.6	53.0	45.5
		Lag 2	37.5	32.2	45.3	32.5	54.8	44.6
		Lag 0	41.8	37.9	43.7	29.8	55.3	43.3

Note: pos. represents improvement and neg. represents deterioration. Moreover, the columns do not add up to 100% as the forecast added value can also be zero. These forecasts are excluded from this table.

6.1. Method

We recruited a sample of 448 participants on MTurk[1]. Participants were randomly assigned to one of two experimental conditions: a traditional forecasting condition and a light-touch forecasting condition. We excluded participants who failed an initial attention check (final $N = 436$): 165 female and 271 male, $M_{age} = 37.2$, $SD_{age} = 10.5$. Participants were ranked on their forecasting performance (by comparing their final forecasts with the products' actual demand). The five best-performing participants were awarded a bonus of $5 on top of the fixed payment of $0.70.

To ensure that the results of our online experiment would be generalized to planning professionals, we conducted the same experiment on Planning Inspiration Day 2019, as organized by the consultancy firm EyeOn. The Planning Inspiration Day is a biannual conference on the future of planning attended by numerous planning professionals. For the 2019 event, we had the opportunity to ask planning professionals in a newsletter before the event to participate in our experiment. Thirty-six planning professionals participated: ten female and 26 male, $M_{age} = 40.7$, $SD_{age} = 9.4$. The best-performing planning professional who participated in the experiment was awarded a gift voucher of €20.

[1] Amazon Mechanical Turk (MTurk) is an online platform used to recruit the participants. MTurk is shown to be a reliable platform to execute online experiments in the behavioral operations literature (Lee et al., 2018).

Participants were randomly assigned to one of two experimental conditions: the traditional forecasting condition and the light-touch forecasting condition. Participants were subsequently told that they would play the role of a sales officer working for a large company and that their goal was to forecast sales for various products. The forecasts used in the experiment were actual forecasts of Company B. For each product, a participant received two different sales forecasts: one forecast generated by an algorithm named SalesCast (representing the statistical forecast) and another forecast generated by the company's marketing department (representing the non-historical information as described in Section 4.1). More details on how the marketing department's forecast was generated are described in Appendix C.

The participant's task was to forecast the future sales of 20 products for the upcoming month (December 2019) after learning the forecasts of both SalesCast and the marketing department. Participants subsequently had to decide for each product whether the forecast of SalesCast should be adjusted, and if so, upwards or downwards (step 1). In case of an adjustment, they also decided on the magnitude of the adjustment (step 2). The traditional forecasting group participants decided on the magnitude of upward and downward adjustments. In contrast, participants in the light-touch forecasting group only decided on the magnitude of the adjustment when they indicated that the forecast of SalesCast should be adjusted downwards. When participants indicated that SalesCast's forecast should be adjusted upwards, an algorithm decided on the magnitude of the upward adjustment in the light-touch forecasting condition. This newly generated forecast was shown to participants (yet they could not adjust it anymore at this stage of the experiment). Appendix D provides more detail on the screens shown to the participants.

6.2. Results and discussion

The results, as depicted in Table 5, indicate that for the MTurk participants, the percentage of adjustments in the upward- and downward direction is similar under both experimental conditions ($\chi^2(1, 436) = 0.219$, $p = .64$, and $\chi^2(1, 436) = 0.054$, $p = .82$ for the two directions, respectively). This means participants do not show a different adjustment direction pattern when confronted with light-touch forecasting. The task completion time is significantly lower under the light-touch forecasting condition, i.e., the participants finished the overall task 40.9% faster in the light-touch forecasting condition ($t_{434} = 8.00$, $p < .001$).

In addition to the MTurk experiment, we also asked planning professionals to participate. Note that given the small sample size of 36 participants divided over two treatments, we only report descriptive statistics for this sample to observe whether the same patterns would emerge in this sample. As shown in Table 6, professional planners under the light-touch planning condition again completed the overall task much faster than participants in the traditional forecasting condition (38% faster on average). The total number of adjustments seems to be higher under the light-touch forecasting method, but we do not know if this is statistically significant; in both cases, the number of upward adjustments is slightly higher than the number of downward adjustments. Note that it took the planning professionals, on average, longer to complete the planning task than the MTurk participants, which can be due to more consideration given to the forecasts by the planning professionals due to their profession.

Table 5
Experimental results on the decision direction and time of MTurk participants.

	Forecasting condition	
	Traditional	Light-touch
N	204	232
Forecasts adjusted [%]	53	57
- Upward [%]	26	28
- Downward [%]	27	28
*Task completion time (s)*Mean	231.9	137.0
Std. Dev.	126.0	121.4

Table 6
Experimental results on the decision direction and time of Planning professionals.

	Forecasting condition	
	Traditional	Light-touch
N	18	18
Forecast adjusted [%]	40	52
- Upward [%]	21	27
- Downward [%]	19	25
Task completion time (seconds)		
Mean	355.2	220.2
Std. Dev.	206.1	106.1

Note that the sample size in Table 6 is too small to draw reliable and meaningful conclusions on whether observed differences are statistically significant between the two experimental conditions.

Note that the sample size in Table 6 is too small to draw reliable and meaningful conclusions on whether observed differences are statistically significant between the two experimental conditions.

Overall, it can be concluded that people do not behave differently in terms of adjustment direction when confronted with the light-touch forecasting method. In contrast, this novel method is significantly faster. These findings imply that a lower forecast error (as reported in Section 5) could also be obtained when the light-touch forecasting method is implemented in companies, with the added benefit that this method is much faster than the currently used forecasting method.

7. Discussion

We have considered the problem of integrating the knowledge of planners in demand forecasting by identifying their strengths and weaknesses in the forecasting process. We identified three findings from the literature that we confirmed in two case studies in two industries. We found that planners regularly adjust demand forecasts and are reasonably accurate in determining the direction of adjustment and magnitude of the downward adjustment. However, the statistical forecasting algorithm

is more accurate than the planner in determining the magnitude of the upward adjustment. We next implemented the findings in a new way of forecasting, named *light-touch forecasting*, in which a planner decides on the direction of the adjustment and the magnitude of the downward adjustment, and an algorithm decides on the magnitude of the upward adjustment. Light-touch forecasting has three main objectives: increase the accuracy of forecasts, decrease the time needed to perform the forecasting task, and prevent gaming of the forecast method by the human planner. We first tested the light-touch forecasting method on two case studies with different time lags. We found a forecast error that was not statistically different from the statistical forecast and the method of Blattberg and Hoch (1990) but was much more time-efficient. Note that we only tested the performance of the 'light-tough forecasting' approach in two case studies; more research is needed to evaluate whether the results are generalizable to other industries and case studies. Next, we performed an online experiment to test whether planners behave differently when confronted with the new forecasting method. The planners' decisions under the light-touch forecasting condition were compared to the traditional way of forecasting, and we found that participants behaved very similarly in terms of the percentage upward and downward adjustments. However, the forecasting task with light-touch forecasting is significantly faster than the traditional forecasting method. We thus achieve our second objective, and we confirm that the performance gain of the case studies is likely to be achieved when implementing our method in practice. Lastly, we show that restricting a planner (such that gaming the system is impossible) is effective. We note that the experiment's results should be taken carefully as the participants received less information than a planner in a real-life setting. In the planning domain, a demand planning team works together to achieve the best possible forecast. This means that the planner integrates information from the sales department, marketing department, and other functions within the supply chain in the final forecast. The alignment on what information to consider is an iterative process (called demand review) in which the demand planning team evaluates input from these different stakeholders to propose the most accurate forecast. This collaborative planning environment is not simulated in the experiment. Therefore, we propose re-examining these results before implementing them in a company.

This paper has several theoretical and practical contributions. First, the multi-method approach employed in this research, i.e., combining field data with an online experiment, is not common in the judgmental forecasting literature, and it makes the results of our study more robust. We confirm previous findings on the strengths and weaknesses of human planners in two new case studies, and we take the next step by implementing this in a new forecasting method. Light-touch forecasting is comparable in terms of forecast error in relation to the statistical model and the method of Blattberg and Hoch, with the time needed to execute the forecasting process being significantly reduced. This makes our forecasting method interesting for implementation within companies. Furthermore, using the method of Blattberg and Hoch (1990), a planner can adapt her forecast, anticipating that the final forecast will be an average of that forecast and the statistical forecast. This is not possible with the light-touch forecasting method.

Various future research directions are possible based on our findings. First, the analyses are based on two companies in demand forecasting. It would be valuable to study if the same patterns arise in other (related) fields and companies, e.g., forecasting demand for perishable products, leading to more generalizable findings. The overarching concept of identifying the strengths of humans while avoiding their weaknesses can also be applied in other fields, as shown by the work of Fügener et al. (2020) in human-AI collaboration. Identifying the strengths and weaknesses of humans operating under different circumstances can be valuable for improving these individual processes during the current Industry 4.0 revolution. Secondly, it would be interesting to study whether similar results can be obtained with new product introductions or products at the end of life, as we have noticed that a significant portion of our data was on these types of products. When comparable results are obtained with these products, we can expand light-touch forecasting to the full range of demand forecasting.

A third possible future research direction is to focus on the individual characteristics of human planners. In our data sets, no information was available on who made which forecasting decision. We could, therefore, not focus on factors such as work experience, risk aversion, and/or trust in the planning system. It would be valuable to get insight into the individual differences of planners. Obtaining this information could lead to a planner-specific definition of light-touch forecasting, probably leading to a performance gain. Fourth, the two case studies used different methods to adjust statistical forecasts (direct adjustment or add/subtract adjustment). It would be interesting to examine the impact of these types of adjustments on the performance of light-touch forecasting. Fifth, to obtain the actual performance gain of light-touch forecasting, it has to be implemented within a company. However, it takes time to get this approved as different parties in the organization need to give permission for this. One organization that participated in this research is interested in implementing light-touch forecasting. Implementing this in close collaboration with the planners is important as it directly affects their daily work. After implementation, we can also examine whether planners indeed do not behave differently when confronted with light-touch forecasting and whether their decision speed is faster. The other future research discussed above could also be performed after such implementation.

We conclude that we have proposed a novel forecasting method that effectively combines human judgment and statistical algorithms. Light-touch forecasting can bring huge performance and efficiency advantages to companies implementing it. At the same time, there still exist plenty of exciting opportunities to research judgmental forecasting and the collaboration between human planners and decision support systems or artificial intelligence.

CRediT authorship contribution statement

B.B.J.P.J. van der Staak: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **R.J.I. Basten:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **P.P.F.M. van de Calseyde:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **E. Demerouti:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **A.G. de Kok:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2024.04.003.

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